

Multiple Choice

1. **Answer:** B

Options: A) Green Bank Telescope; B) Arecibo Telescope; C) Yevpatoria RT-70 Telescope; D) Effelsberg Telescope

Note: Correct option: B - Arecibo Telescope

2. **Answer:** B

Options: A) Being hit in the head by a bullet.; B) Being hit by a small meteorite.; C) Starvation during global winter caused by a major impact.; D) Driving while intoxicated without wearing seatbelts.

Note: Correct option: B - Being hit by a small meteorite.

3. **Answer:** B

Options: A) The materials they are made of are not normally gaseous in everyday experience.; B) They actually contain a significant fraction of non-gaseous matter.; C) The materials that make up these planets are primarily in the form of a plasma not a gas.; D) Actually it's a great description because these worlds are big and gaseous throughout.

Note: Correct option: B - They actually contain a significant fraction of non-gaseous matter.

4. **Answer:** D

Options: A) hydrogen compounds (H₂O CH₄ NH₃) light gases (H He) metals rocks; B) hydrogen compounds (H₂O CH₄ NH₃) light gases (H He) rocks metals; C) light gases (H He) hydrogen compounds (H₂O CH₄ NH₃) metals rocks; D) light gases (H He) hydrogen compounds (H₂O CH₄ NH₃) rocks metals

Note: Correct option: D - light gases (H He) hydrogen compounds (H₂O CH₄ NH₃) rocks metals

5. **Answer:** D

Options: A) Altair A*; B) Alsephina A*; C) Fomalhaut A*; D) Sagittarius A*

Note: Correct option: D - Sagittarius A*

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The altitude of the NCP is the same as your latitude.'

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'The path of the Sun in the sky throughout a year.'

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'the acceleration of the universe.'

Long Form Response

11. **Answer:** Higher in the sky. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** without a strong magnetic field the solar wind stripped away most of the atmosphere significantly reducing the greenhouse effect and cooling Mars down. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key